AC MINI PROJECT

CLASS B PUSH-PULL AMPLIFIER

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Group: A1 (Project number 8)

Aim: Design a **class-B push-pull amplifier** on a multipurpose PCB and make an arrangement to feed the input and observe the output (also verify your circuit using LT SPICE).

Theory:

A class-B push-pull amplifier works by using two complementary transistors (an NPN and a PNP transistor) to handle each half-cycle of the input signal. This configuration allows efficient amplification by ensuring that each transistor only conducts during its respective half-cycle, which reduces power dissipation when there's no input signal.

Key Points of Operation

1. Biasing at the Threshold of Conduction: In a class-B amplifier, the transistors are biased at the threshold of conduction, meaning they only turn on when the input signal is above or below a certain voltage (close to 0.7V for silicon transistors). This minimal bias reduces the standby power dissipation compared to class-A amplifiers, where transistors are biased "on" continuously.

2. Signal Flow and Half-Cycle Amplification:

a) Positive Half-Cycle: When the input AC signal goes positive, the base of the NPN transistor (Q1) becomes positive relative to its emitter, causing it to turn on and conduct. Q1 allows current to flow from the Vcc power supply, through the transistor, to the load (RL). During this time, the PNP transistor (Q2) is off because its base-emitter junction is reverse-biased. The amplified positive half of the input signal is thus passed to the load resistor (RL) via Q1.

b) Negative Half-Cycle- When the input signal goes negative, the base of the PNP transistor (Q2) becomes negative relative to its emitter, causing it to turn on and conduct. Q2 allows current to flow from the load (RL), through Q2, and into the negative supply (Vee). During this time, the NPN transistor (Q1) is off, as its base-emitter junction is reverse-biased. The amplified negative half of the input signal is passed to the load via Q2.

3. Combining the Half-Cycles at the Output: The load (RL) receives the combined effect of both transistors, resulting in a full amplified waveform. The capacitors (C1, C2, and C3) block any DC offset, allowing only the AC component (amplified signal) to reach the output.

Transistors Q1 (NPN) and Q2 (PNP) operate in a complementary manner, one conducts while the other is off, depending on the polarity of the input signal. By splitting the signal into two halves and amplifying each half independently, they create a full sine wave output. The complementary setup (push-pull configuration) ensures that both positive and negative swings of the input signal are amplifier.

One of the challenges in a class-B amplifier is crossover distortion, which occurs near the zero-crossing point of the input signal when the transistors switch from one to the other.

Crossover Distortion: At the zero-crossing point, there's a brief moment when neither transistor is conducting because they're biased right at the edge of conduction. This results in a small distortion at the output, as the signal briefly "drops out" during the transition.

To mitigate crossover distortion, many designs add a small bias voltage to keep both transistors slightly on, a configuration known as class-AB operation, where each transistor conducts slightly into the other's half-cycle.

Power Efficiency: Class-B amplifiers are much more efficient than class-A amplifiers. Each transistor conducts only half the time, reducing the average power dissipated in the transistors. The theoretical maximum efficiency for a class-B amplifier is around 78.5%, which makes it ideal for applications where power efficiency is critical, such as audio amplification.

Circuit Design:

1. Transistor Connections:

- The collector of the NPN transistor (Q1) connects to the positive supply rail (Vcc).

- The collector of the PNP transistor (Q2) connects to the negative supply rail (Vee).

- The emitters of Q1 and Q2 are connected to each other, forming a junction that drives the load resistor (RL) through coupling capacitor C3.

2. Biasing Network:

- R1 and R4 are placed between the input and the base of each transistor (NPN and PNP, respectively).

- These resistors set the base bias for each transistor, which helps define when they begin to conduct based on the input signal.

3. Emitter Resistors (R2 and R5):

- Connected between each transistor's emitter and the common load junction.

- They stabilize the current and help reduce thermal runaway, balancing the circuit's performance.

4. Coupling Capacitors:

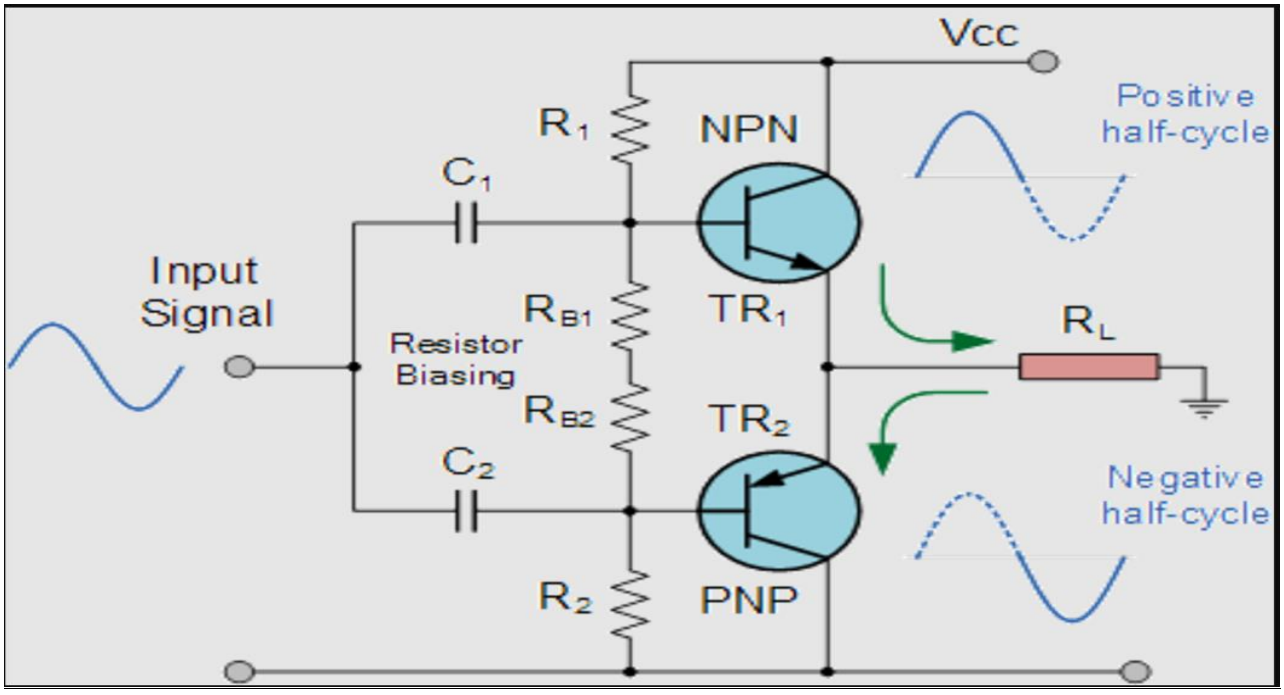
- C1 and C2 are placed in series with the input signal to block any DC offset from the signal source. This prevents any DC bias from affecting the transistors.

- C3 is used at the output to block any DC component from the load, ensuring that only the AC component of the amplified signal is delivered to RL.

5. Output Load (RL):

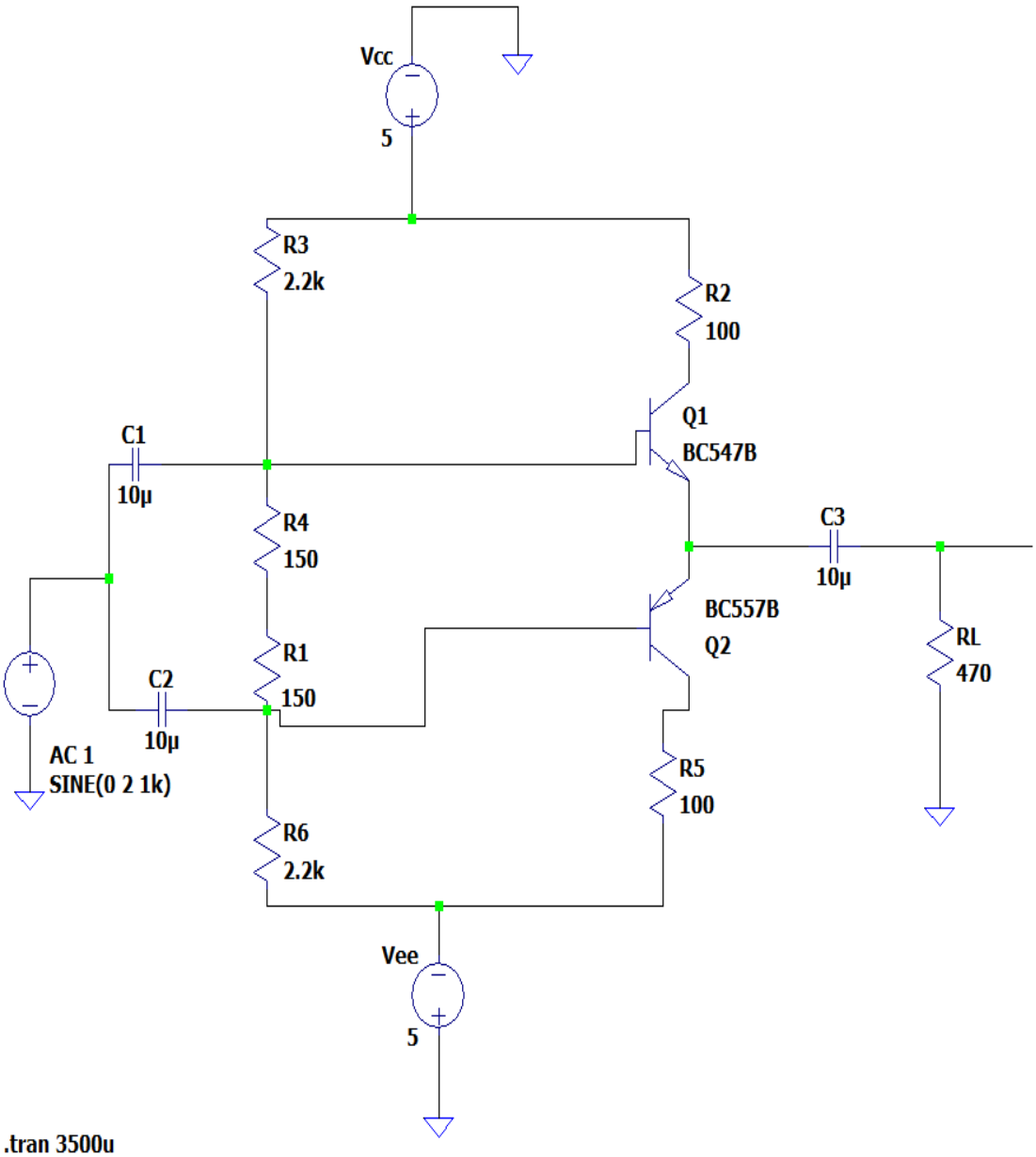
- The output is taken across RL, which is connected in series with C3.

- RL is where the amplified AC signal appears, representing the speaker or other device receiving the output.



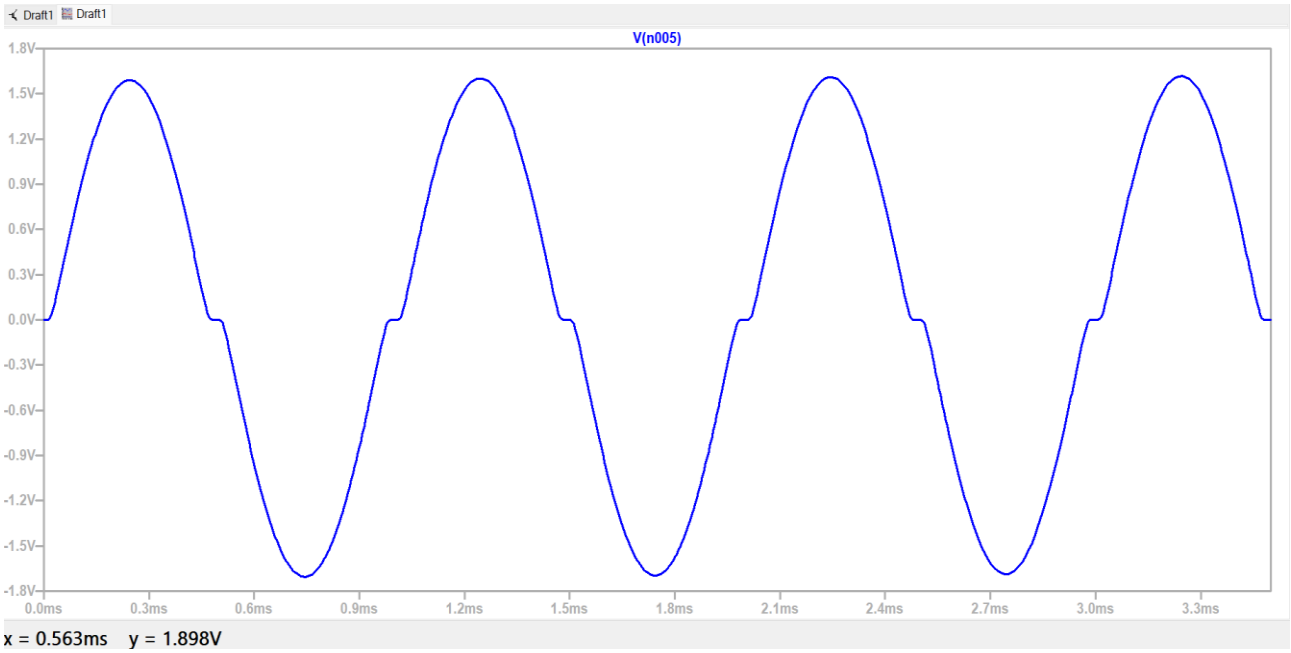
Simulations

1. Software tool used:
   LT Spice
2. Circuit diagram (Load Resistance R_L = 470Ω)

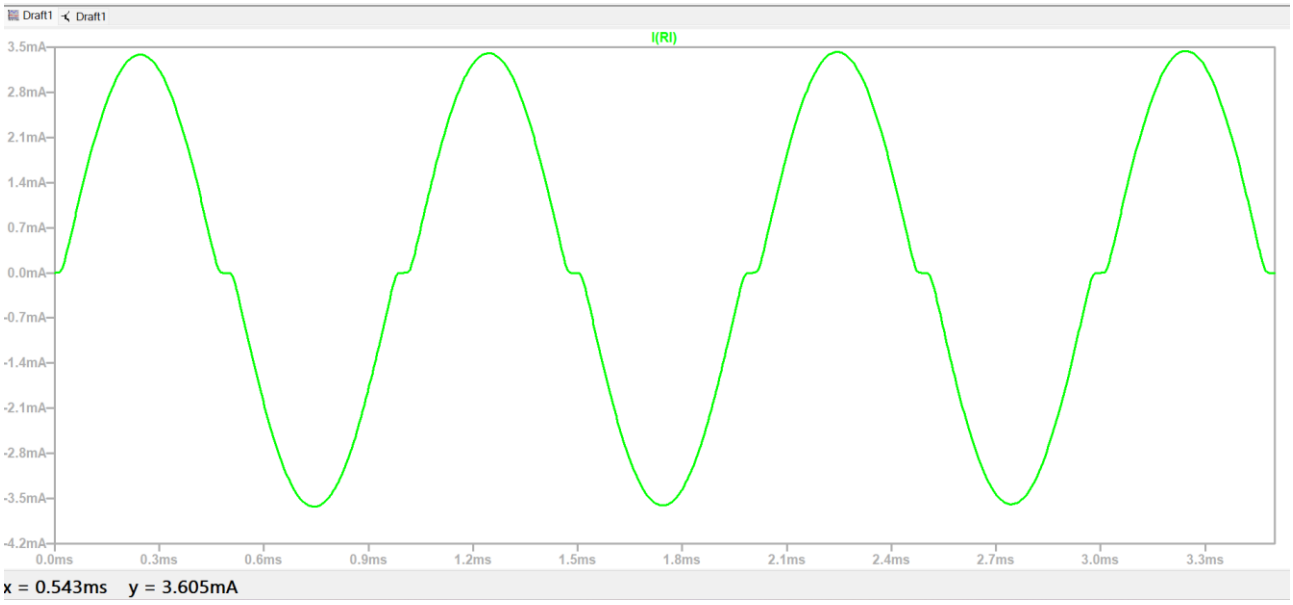


3. Graphs

a) Graph of Vo



b) Graph of current through R_L = 470Ω



4. Results

$P_{out} = V_m I_m/2$ AC Power

$P_{in} = 2V_m I_m/\pi$ DC Power

$\eta = = \pi * V_m * 100/4 * V_{cc}$

Taking the ideal case,

Vpeak = Vcc approximately

$\eta_{max} = \pi/4 =$ **78.5%** (approx)

This gives us the **theoretical efficiency**.

Crossover distortion occurs in class-B amplifiers near the zero-crossing of the input signal when neither transistor conducts due to insufficient base-emitter voltage (\(V_{BE} \approx 0.7V\)). This creates a "dead zone," causing distortion in the output waveform. It can be reduced by applying a small forward bias to the transistors, ensuring they are slightly on even at zero input, effectively converting the amplifier to class-AB operation.

Hardware Experiment

1. Components used:
   - DC Supply (5V)
   - Function Generator
   - Digital Storage Oscilloscope
   - Multipurpose PCB Board
   - Connectors and Headers
   - Soldering wire and iron
   - Resistors (100Ω, 2.2kΩ, 150Ω, 470Ω)
   - Capacitors (10μF)
   - BJT (BC547B NPN, BC557B PNP)

2. Results and Calculation:

   Efficiency (η) = $\frac{Pout(AC)}{Pin\ (DC)}$

   $P_{out} = V_m I_m/2$

   $P_{in} = 2V_m I_m/\pi$

   $V_m = V_{pp}/2 = 400$ mV

   $I_m = 3.605/2$ mA

   $R_L = 470\Omega$

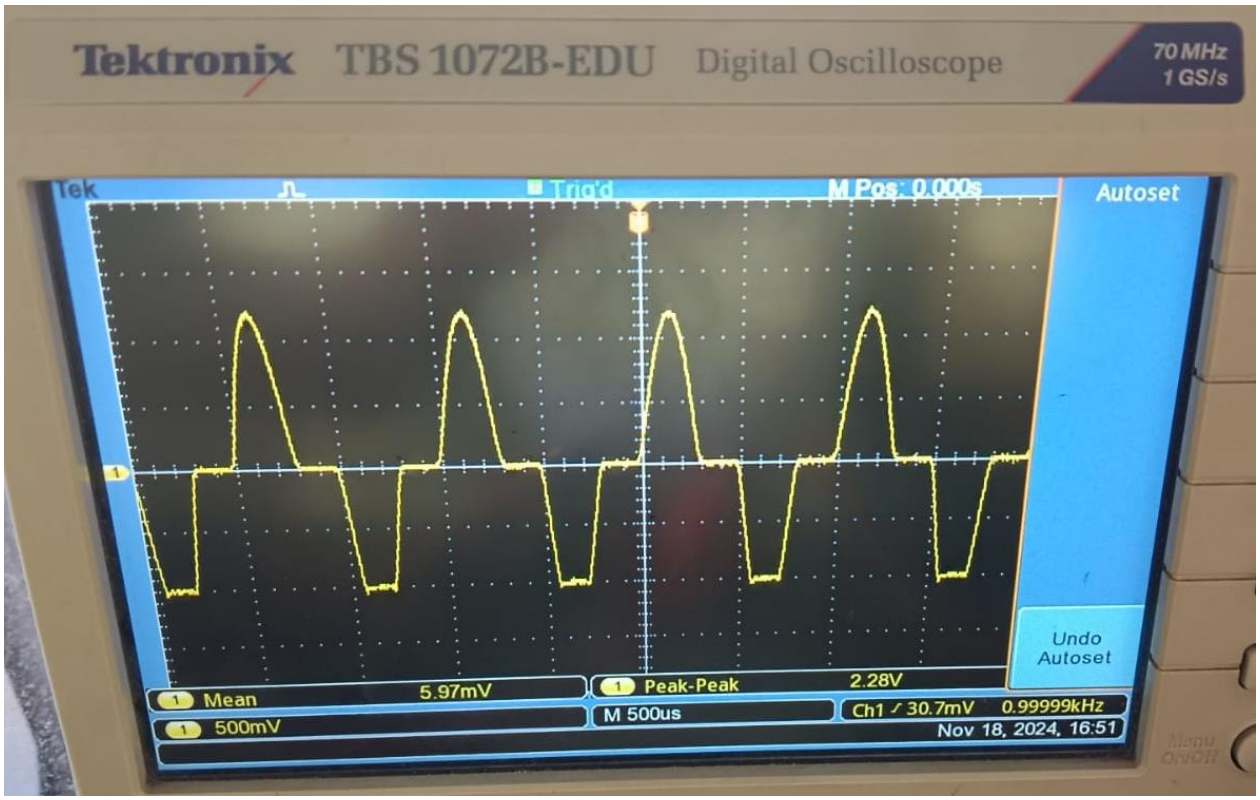
   $P_{out} = 360.5$ mW

   $P_{in} = 5.73$ mW

   $\eta = \pi * V_m * 100/4 * V_{cc}$

   Efficiency (η) = 62.9%

3. Graph obtained on DSO in lab



Conclusion:

In this experiment, a class-B push-pull amplifier was designed, implemented, and analyzed. The amplifier demonstrated high efficiency, with a maximum theoretical value of 78.5%. However, crossover distortion was observed near the zero-crossing point of the input signal due to the transistors' dead zone. Introducing a slight forward bias can reduce this distortion, improving the output waveform quality. The experiment provided insights into the operation, efficiency, and practical challenges of class-B amplifiers, highlighting the trade-offs between efficiency and linearity.